

Data Definition Language (DDL) Commands

Q. Explain the various Data Definition Language (DDL) commands in SQL. Provide the syntax and a clear example for CREATE, ALTER, DROP, TRUNCATE, and RENAME.

[DEF] Definition to Remember

Data Definition Language (DDL) is the subset of SQL used to define, modify, and delete the structural elements (schema) of a database -- such as tables, views, and indexes. DDL commands are auto-committed, meaning changes are saved permanently and cannot be rolled back.

DDL Commands

1. CREATE

Used to

2. ALTER

Used to

3. DROP

Used to

[NOTE] *Once dropped, the table cannot be recovered without a backup.*

4. TRUNCATE

Used to

5. RENAME

Used to

Summary Table

Command	Affects	Data Lost	Auto-Committed
CREATE	Creates table structure	N/A	Yes
ALTER	Modifies table structure	No	Yes
DROP	Deletes table + all data	**Yes (completely)**	Yes
TRUNCATE	Deletes all rows	**Yes (data only)**	Yes
RENAME	Renames table/column	No	Yes

[MUST-WRITE] Must-Write Points (for 10 marks)

1. DDL commands define the schema (structure) of a database.

2. All DDL commands are auto-committed -- changes cannot be rolled back.
3. CREATE: creates a new table/database with specified columns and constraints.
4. ALTER: modifies existing table -- add/drop/modify columns without losing data.
5. DROP: permanently deletes the entire table structure AND all its data.
6. TRUNCATE: deletes all rows but preserves the table structure; faster than DELETE.
7. DROP removes structure + data; TRUNCATE removes data only (structure remains).

[RECALL] Quick Recall

DDL -> Auto-committed -> CREATE (make table) -> ALTER (modify structure) -> DROP (delete all) -> TRUNCATE (delete rows, keep structure) -> RENAME (change name)